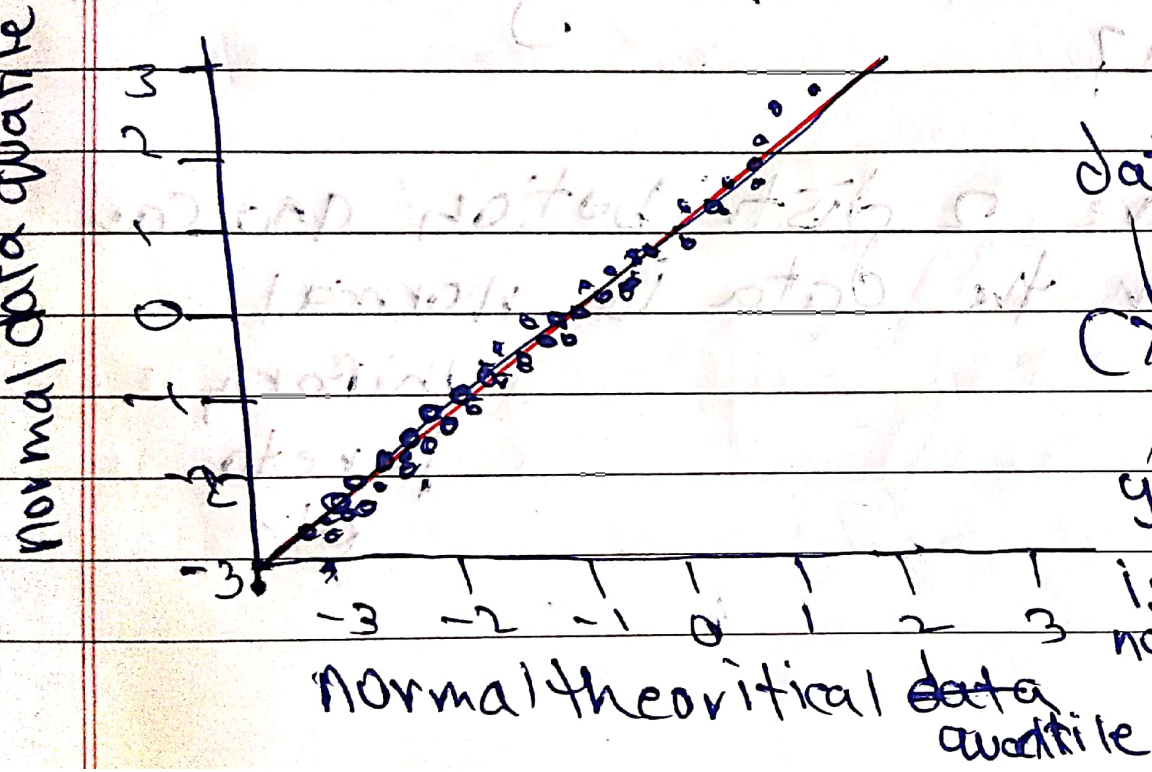
To find weather a ~~plot~~ data is normal we use many method, one of the efficient method is

QQ Plot : A QQ Plot (quantile-quantile plot) is a graphical tool used to assess the similarity of the distribution of 2 sets of data. It is particularly useful for determining wheather a set of data follows normal distribution.



data

(X, Y)

yes → is normal

no → not normal

theoritical distribution

This shows how our is distributed and compared without data X

Steps:

i) take theoretical (y) → first sort data → then find out its quantile

1st
2nd
100nd

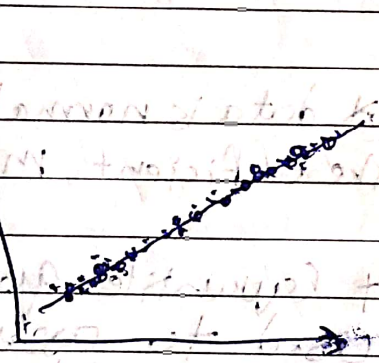
y - quantile

ii) take your data (x) → first sort → same find its quantile

x_quantile

iii) (x_quantile, y_quantile) and plot it

if



if every points come on single line segment then its normal distributions

Is this distribution used only for normal distribution?

No it compare 2 distribution and can tell weather the data is normal
Uniform
Pareto
etc.